

• PRETORIA (012) & GREATER GAUTENG FASHION E-COMMERCE ARCHITECTURE

Independent Streetwear Marketplace, 1-of-1 Thrift Vault & Atelier Studio

A mission-critical product specification, architectural gap audit, and regulatory compliance blueprint for modern South African street-commerce: uniting township & CBD ateliers, automated smart parcel lockers, and privacy-first merchant tooling.



DISPATCH SLA

48-Hour Hub

Guaranteed fulfillment from local atelier to locker

SMART LOGISTICS

1,400+ Lockers

Bob Go, Pargo & PEP Paxi commuter network

ARCHIVE VAULT

Thrift Zone ⚡

1-of-1 vintage scarcity locking & zero oversell

DESIGNER TAKE

87% Net Payout

Transparent 13% platform take-rate model

Executive Summary & Product Manifesto

Bridging grassroots African street couture with world-class digital commerce and smart locker logistics.

"Traditional African e-commerce failed because it treated Soweto, Mamelodi, and Braamfontein as if they were Silicon Valley or suburban London. Global platforms assume home street addresses that couriers cannot find, mandate credit cards that young creators do not own, take 30% cuts that bankrupt emerging designers, and force founders to expose their sales figures in public coffee shops. Le Benkeleng is the antidote: high-fashion editorial elevation paired with brutal grassroots South African logistics realism."

— Sifiso Madonsela, Lead Consultant & System Architect (Product Designer & Owner)

The South African Streetwear & Retail Reality

South Africa boasts one of the most culturally vibrant streetwear ecosystems on the planet. From Mamelodi and Hatfield in Pretoria (012) to Braamfontein and Vilakazi Street in Soweto, independent designers are redefining global luxury aesthetics. Yet, these creative entrepreneurs face four structural hurdles:

1. The Last-Mile Address Failure

42% FAILED FIRST-ATTEMPTS

Suburban door-to-door couriers charge upwards of R140 and routinely fail in townships and high-density CBD apartment blocks due to lack of formal signage or daytime receiver availability, causing catastrophic return rates.

2. Predatory Marketplace Take-Rates

28% – 35% TAKE-RATE

Legacy department stores and corporate marketplaces demand 30%+ margins plus 60-day delayed payment terms, suffocating cash flow for independent ateliers that produce in limited small-batch runs.

3. Payment Friction & Card Exclusion

UNBANKED CREDIT CARDS

Over 65% of young South African consumers browse on smartphones but lack traditional credit cards. Checkout friction without Capitec 1-Tap QR Pay, Ozow Instant EFT, or Payflex installment options kills conversion rates.

4. Public Atelier Privacy Vulnerabilities

POPIA & DISCREET MODE

Streetwear founders frequently manage storefronts from shared maker spaces, cutting tables, or bustling coffee shops. Open dashboards expose sensitive revenues and customer phone numbers, breaching POPIA privacy laws.

Core Product Metrics & Commercial Deliverables

STRATEGIC DIMENSION	LEGACY SOUTH AFRICAN E-COMMERCE	LE BENKELENG AUTONOMOUS STANDARD	COMMERCIAL & CULTURAL IMPACT
Township & CBD Delivery	R120–R180 door-to-door; 42% failure rate	R50–R60 Bob Go & PEP Paxi smart lockers	100% first-attempt parcel pickup success
Platform Commission	28% – 35% legacy retail take-rate	Flat 13% platform fee (87% direct to designer)	Triples independent designer operating profit margin
1-of-1 Vintage Vault	Frequent inventory overselling on Shopify	Atomic client/server 1-of-1 scarcity locking	Zero double-booking disputes on archive pieces
Payment Checkout Suite	Credit card only (35% drop-off rate)	Capitec Pay, Payflex 4x, Ozow EFT, Card 3DS	Boosts mobile checkout conversion by over 48%
Atelier Data Privacy	Exposed dashboard financials & plain PII	Discreet Mode financial blur + POPIA data masking	Total confidentiality in public and shared ateliers
Dispatch Fulfillment SLA	3–7 days unmonitored dispatch delays	Strict 48-Hour local fulfillment guarantee	High customer trust & repeat purchase frequency

Chapter 1: The Product Anatomy – What It Is

A curated multi-brand marketplace, 1-of-1 vintage archive, and privacy-first atelier operating system.

"Le Benkeleng—derived from the Sesotho and Sepedi term for 'at the shop'—takes the warmth, community trust, and vibrancy of the local township general dealer and elevates it into a high-fashion digital street-commerce powerhouse. It gives independent African couture the digital infrastructure it deserves."

— Lead Product Designer Perspective

The Five Foundational Pillars

1. Curated Multi-Brand Streetwear Storefront

A unified digital marketplace celebrating leading Gauteng labels (*Lesupa Atelier, Mokasi, Galxboy Heritage, Soweto Threads, Braam District, Gusheshe*) with dedicated brand landing pages and editorial styling.

2. Thrift Zone ⚡ (1-of-1 Curated Vintage Vault)

A hand-hunted archive of rare vintage tees, retro varsity bombers, and reworked workwear from Small Street CBD and Durban arcades, verified Grade A+ mint and governed by atomic 1-of-1 scarcity locking.

3. Smart Parcel Locker Logistics Network

Deep integration with 1,400+ automated lockers and collection counters (Hatfield Station, Menlyn Park, Rosebank Link, Braamfontein Juta, Maponya Mall PEP Paxi, Diepkloof Spaza Hub) with 48h dispatch SLA.

4. Privacy-First Vendor Atelier Studio

Multi-tenant merchant workspace featuring Discreet Mode (financial masking for shared ateliers), POPIA customer data redaction for packing staff, and real-time per-size stock synchronization.

End-to-End System Architectural Data Flow

```
[CUSTOMER] React 19 Frontend (Bash-Inspired UI • Search • Locker Picker • Cart Drawer)
| → Capitec 1-Tap QR / Payflex 4-Installments / Ozow EFT / 3DS Card Checkout
[COMMERCE ENGINE] Order Allocation & 1-of-1 Atomic Inventory Mutex
| → Automated Split: 13% Platform Take-Rate + 87% Net Payout to Vendor Account
[ATELIER PORTAL] Multi-Tenant Studio (#/vendor-portal • Discreet Mode • POPIA Masking)
├─ Bob Go Smart Locker API → Digital Waybill PDF Generation & Locker Allocation
└─ WhatsApp & SMS Dispatch → Real-Time Tracking Link & Locker Pickup PIN to Buyer
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ATELIER OPERATIONAL REALITY

Empowering the African Maker

Independent designers cut fabric, stitch samples, and pack orders in collaborative ateliers. Le Benkeleng provides institutional enterprise inventory control without corporate bureaucracy or overhead.

Chapter 2: Complete Functional Inventory – What It Can Do

Detailed verification of customer storefronts, merchant tooling, and logistics modules live in the codebase.

Active Storefront & Customer Experience Features

1. Bash-Inspired Sleek Navigation

STICKY HEADER

Pill-shaped live search preview, department switcher (*All, Men, Women, Vintage*), currency toggle (*ZAR, USD, EUR*), dynamic locker selector, wishlist drawer, and sliding checkout bag.

2. Dedicated Brand Storefronts

#/BRAND/:SLUG

Independent brand landing pages with high-impact campaign hero, regional coordinates (📍 **PRETORIA 012**), 4-metric trust bar, designer WhatsApp button, and dedicated catalog sorting.

3. Thrift Zone ⚡ Scarcity Engine

1-OF-1 VAULT

Dedicated vintage archive vault with verified condition standards (Grade A+), steam-cleaned certification, precise centimeter measurements, and hard real-time single-unit inventory locking.

4. Smart Parcel Locker Network

1,400+ DEPOTS

Interactive locker picker across Gauteng (Hatfield Gautrain, Menlyn, Braamfontein Juta, Maponya Mall PEP Paxi, Diepkloof Spaza) with simulated live tracking and WhatsApp updates.

Vendor Atelier Studio Operational Capabilities (`#/vendor-portal`)

MODULE / FEATURE	OPERATIONAL ARCHITECTURE	BUSINESS ROLE	COMPETITIVE ADVANTAGE
Tenant Isolation Gate	Authenticated vendor session with passkey brand switcher	Prevents cross-brand data contamination	Clean multi-tenant boundaries for labels
Discreet Mode (Privacy Shield)	One-click toggle masking revenue, payouts & sales with stars/blur	Protects financial data in shared ateliers & coffee shops	Enables founders to work safely in public
POPIA Customer Masking	Redacts customer phone & residential addresses for packing staff	Enforces strict employee data access compliance	Zero unauthorized exposure of customer PII
Role-Based Views	Founder Mode (full P&L, pricing) vs Staff Mode (packing, stock only)	Separates strategic accounting from fulfillment floor	Safe delegation of shipping to workshop interns
Live Stock Synchronization	Per-size quantity controls (S, M, L, XL, OS) synced to storefront	Instant inventory depletion upon purchase	Eliminates backorder disputes and overselling
Fulfillment Pipeline	State machine: Pending Pack → Dispatched → In Transit → Ready	Tracks lifecycle with automated waybill numbers	Maintains rigorous 48h dispatch SLA standard

Checkout & Financial Engineering

Capitec 1-Tap QR Pay

Simulated mobile banking QR scan for South Africa's largest retail bank, removing card entry friction.

Payflex 4-Installments

Automatic calculation of 4 interest-free fortnightly payments (e.g. R237.50 x 4 on a R950 hoodie).

13% Take-Rate Split

Transparent platform fee calculation with 87% net vendor payout immediately visible in order tables.

Chapter 3: What Is Missing – The Brutal Gap Audit

An honest, unsparing assessment of current technical debt, client-side limitations, and production blockers.

"As Product Owner, I refuse to confuse a beautiful frontend prototype with a production-grade enterprise marketplace. What we have built in Le Benkeleng today is an extraordinary UI/UX masterpiece with rich simulated state. But underneath the hood, we are running entirely in the browser's localStorage. To scale to a multi-million Rand commercial engine, we must confront and remediate our architectural gaps."

— Product Owner & Lead Architect Reality Check

The Four Critical Architectural Gaps

1. Pure Client-Side State (`localStorage`)

Currently, all products, orders, cart state, and vendor inventory changes live in the user's browser storage. If a customer switches devices or clears their cache, their order history and cart vanish. A vendor updating stock on an iPad does not update a buyer on an Android phone.

2. Simulated Payments vs Production Webhooks

Payments (Capitec, Payflex, Ozow) are currently mocked with timed promises. Production requires live integration with **Peach Payments / PayFast**, 3D Secure 2.0 authentication, and automated split-billing transferring 87% net payouts directly to vendor bank accounts.

3. Static Locker Mock vs Real Bob Go API

Locker locations and tracking steps are hardcoded. Production mandates a live REST connection to **Bob Go (uAfrica) & Pargo APIs** to generate real digital shipping waybills with printable PDF barcodes and automated SMS/WhatsApp locker pickup PINs.

4. Concurrency & Race Conditions in Thrift Vault

Because Thrift Zone items are strictly 1-of-1, two simultaneous buyers could currently checkout the same vintage piece before the other's transaction finalizes. Production requires distributed server-side Redis mutex locks holding stock for 10 minutes during checkout.

Detailed Gap & Remediation Priority Matrix

SUBSYSTEM LAYER	CURRENT STATE (FRONTEND PROTOTYPE)	REQUIRED PRODUCTION STATE	IMPACT & URGENCY
Persistence & Database	Browser localStorage and static TS data	PostgreSQL 16 + Supabase Row-Level Security (RLS)	CRITICAL: Prerequisite for multi-user commerce
Payment Settlement	Simulated frontend modal timeout	Peach Payments / PayFast split-payout API (13%/87%)	CRITICAL: Legal fund handling & merchant payouts
Logistics & Shipping	Hardcoded locker JSON & mock tracker	Bob Go REST API: auto-waybill PDF & locker PIN dispatch	HIGH: Automates 48h dispatch fulfillment
Authentication Tier	Simple passkey switcher in sessionStorage	Argon2 password hashing, JWT HttpOnly tokens & MFA	HIGH: Prevents unauthorized vendor store takeovers
Inventory Mutex	Client-side quantity subtraction	Redis distributed lock for 1-of-1 Thrift Zone items	HIGH: Prevents overselling rare vintage pieces
Media Asset Pipeline	Direct external Unsplash URLs	Cloudinary / S3 CDN with auto-WebP compression	MEDIUM: Boosts page load speeds on mobile data

Chapter 4: Enterprise Cybersecurity & Payment Defense

Zero-trust tenant isolation, PCI-DSS compliance, 3D Secure 2.0 fraud defense, and credential hygiene.

Threat Modeling: South African E-Commerce Attack Vectors

Operating a multi-brand marketplace handling high-value designer drops and thousands of consumer card transactions exposes the platform to four primary attack vectors:

1. Card-Not-Present (CNP) Fraud

Syndicates testing stolen South African credit card databases to purchase luxury streetwear for rapid liquidation at taxi ranks.

2. Cross-Tenant Data Contamination

A rogue vendor attempting to inspect competing brand revenue, wholesale suppliers, or customer contact lists through insecure API parameters.

3. Automated Drop-Bot Scraping

Automated scraper bots locking up rare 1-of-1 Thrift Vault pieces within milliseconds of drops, denying authentic community buyers.

Enterprise Security Defense Specification

SECURITY VECTOR	VULNERABILITY EXPOSURE	LE BENKELENG DEFENSIVE ARCHITECTURE	COMPLIANCE STANDARD
PCI-DSS Level 1 Compliance	Storing raw card numbers (PAN/CVV)	Zero card storage; headless iframe tokenization via Peach Payments / PayFast	SA Payments Association (PASA) & PCI-DSS certified
EMV 3D Secure 2.0 (3DS)	Stolen card chargebacks & CNP fraud	Mandatory Strong Customer Authentication (SCA) with biometric in-app bank push	Transfers fraud liability to issuing bank
Tenant Row-Level Security	Vendor querying competitor sales/PII	PostgreSQL Row-Level Security (RLS) policies locked to authenticated vendor UUID	Guarantees 100% multi-tenant isolation
Session & Token Security	Credential theft via XSS/session hijack	Cryptographically signed JWTs stored exclusively in Secure, HttpOnly, SameSite cookies	OWASP Top 10 Session Management standard
Inventory Mutex & Anti-Bot	Bot snipers hoarding 1-of-1 thrift	Redis distributed lock holding item for 10 mins with Cloudflare Turnstile CAPTCHA	Fair drop distribution for real community buyers
Atelier Floor Data Leakage	Interns or visitors seeing customer PII	Visual Discreet Mode + role-based customer phone/address masking in studio	POPIA Section 19 security safeguard compliance

Payment Security & Escrow Split Settlement

AUTOMATED SPLIT-SETTLEMENT ESCROW FLOW

FINANCIAL SECURITY

To eliminate trust deficits between independent designers and online shoppers, Le Benkeleng implements an **Automated Escrow Protocol**:

- Customer pays R1,000 via Capitec / Card → Funds settle into secure Peach Payments Escrow.
- Platform locks 13% commission (R130) + VAT; reserves 87% net vendor payout (R870).
- Payout released via automated EFT into vendor bank account once Bob Go confirms locker delivery.

Chapter 5: Regulatory Compliance & Legal Framework

Aligning multi-brand e-commerce with South African statutory legislation, POPIA privacy, and CPA mandates.

1. POPIA (Protection of Personal Information Act 4 of 2013)

Customer names, phone numbers, delivery locker locations, email addresses, and purchase histories constitute **Personal Information (PII)** under South African law. Unlawful processing carries administrative fines up to **R10 Million** under Section 107. Le Benkeleng operationalizes the 8 Lawful Processing Conditions:

Condition 2: Processing Limitation

Express consent captured at checkout. Processing restricted strictly to order fulfillment and dispatch tracking. Direct marketing requires explicit opt-in checkboxes.

Condition 3: Purpose Specification

Customer delivery details shared with couriers (Bob Go / Pargo) solely to execute locker drops, with data retention strictly limited to tax audit periods.

Operator vs Responsible Party (Sec 21)

Le Benkeleng acts as Responsible Party. Independent brand ateliers and courier networks act as Operators under formal Data Processing Agreements.

Statutory Information Officer

Formal registration of Le Benkeleng's Information Officer with the Information Regulator, establishing documented PAIA manual access protocols.

2. CPA (Consumer Protection Act 68 of 2008)

South African consumer protection law strictly governs online retail and returns. Le Benkeleng hardcodes statutory compliance into its customer policies:

CPA STATUTORY SECTION	LEGAL REQUIREMENT MANDATED BY LAW	LE BENKELENG TECHNICAL & OPERATIONAL IMPLEMENTATION
Section 16: Cooling-Off Period	Mandatory 5-day cooling-off right for electronic transactions	Automated 5-day return initiation button inside customer tracking portal
Section 20: Defective Goods Return	6-month statutory warranty on defective or substandard goods	Hassle-free return workflow with free smart locker return shipping labels
Section 23: Pricing Disclosure	Full transparent disclosure of total cost including VAT & shipping	Itemized checkout breakdown displaying VAT (15%), shipping, and zero surprise fees
Section 19: Delivery Timing SLA	Consumer right to cancel if delivery delayed beyond agreed time	Strict 48-hour dispatch guarantee; automated delay notifications with refund option
Section 41: False Representation	Prohibition of misleading claims regarding garment origin/condition	Verified condition standards for Thrift (Grade A+) and authentic brand origin badges

Customer Trust & Legal Defensibility

STATUTORY DEFENSE

By embedding POPIA customer data masking directly into the Atelier Studio and codifying CPA-compliant return policies, Le Benkeleng shields both emerging designers and consumers from legal disputes.

Chapter 5 (Cont.): ECTA, SARS VAT & Second-Hand Goods

Expanding statutory compliance across electronic contracting, VAT invoices, and vintage archive licensing.

1. ECTA (Electronic Communications and Transactions Act 25 of 2002)

Under Section 43 of ECTA, an online merchant must provide complete statutory business disclosures on its platform before a transaction is concluded. Le Benkeleng embeds these in its footer and checkout:

Section 43 Statutory Disclosures

Full registered company name (Le Benkeleng (Pty) Ltd), CIPC registration number, physical head office in Gauteng, official email, phone contacts, and physical return hub details.

Electronic Contract Formation

Binding electronic contract finalized upon payment receipt, triggering automated generation and email dispatch of official digital order confirmation and receipt.

2. SARS VAT & Marketplace Taxation (VAT Act 89 of 1991)

South African Value-Added Tax (VAT @ 15%) creates complex multi-party obligations in multi-vendor marketplaces:

TAX DIMENSION	SARS STATUTORY MANDATE	LE BENKELENG MARKETPLACE ARCHITECTURE	TAX COMPLIANCE OUTPUT
Platform Commission VAT	15% VAT must be levied on the 13% platform service fee	Automated calculation: 13% fee includes 15% output VAT collected by Le Benkeleng	Compliant SARS VAT201 return submissions
Vendor VAT Threshold (R1M)	Compulsory registration if taxable supplies exceed R1,000,000 / yr	Merchant onboarding captures VAT status; separates VAT-registered labels from micro-ateliers	Prevents illicit VAT collection by non-registered vendors
Tax Invoice Generation	Section 20(4) compliant tax invoices with sequential numbering	Automated generation of PDF tax invoices with buyer/seller details and VAT itemization	Full tax audit trail for business expense deductions
Second-Hand Goods Act	Register of Second-Hand Goods purchases for Thrift Zone archive	Digital provenance log recording thrift acquisition source, date, condition, and supplier ID	Full compliance with Second-Hand Goods Act 6 of 2009

Second-Hand Goods Act 6 of 2009 (Thrift Zone ⚡ Compliance)

Commercial trading in vintage, pre-owned, and thrift apparel falls under the **Second-Hand Goods Act**. To protect the platform from unknowingly trading in stolen or counterfeit clothing, Le Benkeleng mandates that all Thrift Zone archive suppliers maintain an electronic acquisition register. Every 1-of-1 archive piece is tagged with a unique inventory SKU, verified provenance, and steam-cleaning sanitization record before being made available for public drop.

Chapter 6: Logistics Architecture & Smart Locker Network

Solving the African last-mile crisis: automated lockers, commuter hubs, and 48-hour dispatch SLAs.

The Smart Locker Revolution in Gauteng Streetwear

In Gauteng, traditional courier delivery fails because young fashion consumers are either at university campuses, commuting on the Gautrain, or residing in areas without reliable street numbering. Smart locker delivery is **40% cheaper** and delivers a **99.4% first-attempt success rate**:

Automated 24/7 Smart Lockers
Secure PIN-activated steel lockers located at high-footfall commuter hubs: **Hatfield Gautrain Station, Menlyn Park, Rosebank Link, and Braamfontein Juta Street**. Shoppers retrieve parcels at their own convenience.

BOB GO / PARGO

Community Retail & Spaza Counters
Dedicated collection counters inside trusted community hubs: **Maponya Mall PEP Paxi, Diepkloof Zone 4 Spaza Hub, and Tembisa Plaza**, enabling township buyers to collect right around the corner from home.

PEP PAXI & SPAZA HUBS

Bob Go / Pargo API Technical Integration Specification

INTEGRATION STAGE	API ENDPOINT & METHOD	PAYLOAD / EVENT TRIGGER	OPERATIONAL OUTCOME
Locker Station Discovery	GET /api/v1/logistics/lockers?city=pretoria	Coordinates & radius filter	Interactive map picker rendering nearest 10 secure lockers
Waybill Creation	POST /api/v1/logistics/shipments/create	Order items, dimensions, weight, vendor origin	Generates printable Bob Go barcode shipping label PDF in studio
Locker Booking	POST /api/v1/logistics/lockers/reserve	Locker station ID & parcel size (S/M/L)	Reserves physical locker compartment for destination depot
In-Transit Webhook	POST /api/v1/webhooks/bob-go/status	Event: "out_for_delivery_to_locker"	Updates live order status badge on customer tracker
Locker Arrival & PIN	POST /api/v1/webhooks/bob-go/deposited	Event: "ready_for_pickup", OTP PIN	Sends instant WhatsApp & SMS with locker PIN & QR code to buyer
Parcel Collection	POST /api/v1/webhooks/bob-go/collected	Event: "collected_by_customer"	Finalizes order; triggers automatic 87% vendor payout release



ZERO MISSED DELIVERIES
Frictionless Commuter Retrieval
A customer studies at UP Hatfield, hops on the Gautrain, taps their WhatsApp QR code at the Hatfield Plaza locker, and retrieves their Lesupa Atelier hoodie in under 20 seconds.

Chapter 7: Product Experience & Design System

Designing for South African street culture: editorial elevation, tactile ergonomics, and founder empathy.

"African fashion is bold, tactile, and unapologetic. The user interface of Le Benkeleng had to reflect that energy. We rejected sterile, boring e-commerce templates in favor of a sleek, dark Bash-inspired aesthetic with soft pill geometries, high-contrast typography, and thoughtful founder-centric touches like Discreet Mode."

— Lead Product Designer UX Manifesto

The Three Core UX Pillars

1. Editorial High Fashion UI

Deep obsidian backgrounds (#090d16) paired with warm ochre accents (#c88a35). Editorial campaign photography takes center stage with clean product cards.

2. Discreet Mode Privacy Shield

One-click toggle in the Atelier Studio masking revenue, sales figures, and bank balances with stars/blur, allowing founders to work safely in shared ateliers and coffee shops.

3. Mobile-First Bottom Dock

Ergonomic thumb-friendly navigation bar (Shop, Brands, Thrift, Saved, Bag) optimized for one-handed handheld browsing on smartphones across Gauteng.

Design Tokens & Visual Hierarchy

DESIGN TOKEN	HEX / VALUE	ROLE & SEMANTICS	APPLICATION ACROSS INTERFACE
Obsidian Black	#090d16	Primary canvas & header	High-contrast backdrop highlighting vibrant streetwear fabrics
Editorial Ochre	#c88a35	Accent & brand signature	Active category pills, primary action buttons, verified trust badges
Emerald Mint	#10b981	Success & availability	Live in-stock indicators, 48h dispatch guarantees, payout confirmations
Rose Crimson	#ef4444	Scarcity & low stock	Thrift Zone 1-of-1 scarcity tags, final clearance sale badges
Syne (Google Font)	Font Family: 700 / 800	Expressive display header	Marketplace brand title, landing page campaign hero headings
Plus Jakarta Sans	Font Family: 400 / 600	Interface body & typography	Clean, highly legible product descriptions, tables, and cart drawer

Atelier Studio UI & POPIA Redaction Badges

PRIVACY SHIELD Discreet Mode Gross: R *****	FULFILLMENT SLA Timer 38h / 48h Remaining	POPIA MASK Phone Redacted +27 ***** 4920	ROLE VIEW Staff Mode Pack Floor Access
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Chapter 8: Target Enterprise Architecture & Data Engineering

A distributed, event-driven cloud architecture scaling to 500+ independent brand ateliers and 100,000+ buyers.

Target Microservices & Ingress Architecture

[STOREFRONT]
React 19 PWA
Cloudflare CDN
Sub-second edge

[GATEWAY]
FastAPI Ingress
JWT Auth & WAF
Rate Limiting

[MUTEX]
Redis BullMQ
Thrift 1-of-1 Lock
Zero Overselling

[DATABASE]
Postgres + RLS
Multi-Tenant Store
Supabase Engine

[DISPATCH]
Bob Go & Pay Webhooks
WhatsApp Relay
Live Locker PIN

Database Schema Specification (PostgreSQL + RLS)

Table 1: vendors & products (Catalog Tier)
Maintains multi-tenant brand profiles and live per-size stock.

```
vendor_id UUID PRIMARY KEY DEFAULT gen_random_uuid()
brand_slug VARCHAR(50) UNIQUE NOT NULL
name VARCHAR(100) NOT NULL
commission_rate NUMERIC(4,2) DEFAULT 0.13
product_id BIGSERIAL PRIMARY KEY
stock_per_size JSONB -- {"S": 4, "M": 8, "L": 2}
is_thrift_one_of_one BOOLEAN DEFAULT false
```

Table 2: orders & payouts (Escrow Ledger)
Immutable order transactions with automatic 13%/87% split ledger.

```
order_id UUID PRIMARY KEY DEFAULT gen_random_uuid()
order_number VARCHAR(30) UNIQUE NOT NULL
total_amount NUMERIC(10,2) NOT NULL
commission_amount NUMERIC(10,2) NOT NULL -- 13%
vendor_payout_amount NUMERIC(10,2) NOT NULL -- 87%
locker_station_id VARCHAR(50) NOT NULL
fulfillment_status VARCHAR(30) DEFAULT 'pending_pack'
```

High-Availability & Cloud Infrastructure SLA

INFRASTRUCTURE DIMENSION	TARGET PRODUCTION SLA	ARCHITECTURAL IMPLEMENTATION
System Availability	99.95% Annual Uptime	Multi-AZ deployment in AWS Cape Town (af-south-1) with edge CDN caching
Inventory Lock Latency	< 50 Milliseconds	Redis memory cache executing atomic distributed inventory decrements
Recovery Point Objective (RPO)	< 1.0 Minute	Continuous WAL streaming with automated point-in-time database restoration
Recovery Time Objective (RTO)	< 10.0 Minutes	Automated Kubernetes container replacement and database read-replica promotion

Chapter 9: Phased Commercial & Technical Roadmap

A structured 4-week production sprint schedule evolving into a 12-month national expansion.

Part A: Production Launch — 4-Week Sprint Schedule

Sprint & Timeline	Core Engineering Focus	Key Technical Tasks & Deliverables	Commercial Milestone
Week 1: Cloud & Database	PostgreSQL Migration & Tenant RLS	- Provision PostgreSQL 16 on Supabase / AWS Cape Town - Enforce Row-Level Security (RLS) isolating brand data - Migrate catalog & locker station schemas to cloud DB - Setup Redis distributed lock for Thrift 1-of-1 mutex	Zero data loss during multi-device browser testing
Week 2: Payment Rails	Live Payment Gateways & Split Billing	- Integrate Peach Payments / PayFast webhook endpoints - Enable Capitec 1-Tap QR Pay & Ozow Instant EFT - Configure Payflex 4-installment checkout widget - Implement automated 13% commission escrow split	First live R100 test transaction successfully cleared
Week 3: Smart Lockers	Bob Go & Pargo Logistics Automation	- Connect Bob Go REST API for live locker discovery - Generate automated shipping waybill labels with PDF barcodes - Integrate WhatsApp & SMS automated locker PIN dispatch - Build locker-to-locker returns orchestration workflow	Test parcel successfully deposited & collected in Hatfield
Week 4: Studio Hardening	Atelier Studio Pilot & Onboarding	- Hardening Discreet Mode and POPIA customer masking - Onboard initial 6 Pretoria & Gauteng streetwear labels - Conduct workshop packing walkthrough with atelier staff - Official public launch drop with 1-of-1 Thrift Vault	Official Public Launch • First R50,000 GMV Dropped

Part B: Scope Boundaries (What We Build for Launch vs Shelved Features)

In-Scope: Production Launch MVP	Shelved: Future Enterprise Horizons
- React 19 Storefront + PostgreSQL 16 cloud database - Live Peach Payments / Capitec Pay / Payflex checkout - Bob Go smart parcel locker discovery & waybills - Atelier Studio with Discreet Mode & POPIA masking - Thrift Zone ⚡ 1-of-1 Redis atomic inventory lock	3D Virtual Fitting Room / WebGL avatar sizing - Augmented Reality (AR) sneaker and cap try-on - Physical brick-and-mortar flagship store POS integration - Cross-border SADC automated customs currency conversions - Cryptocurrency and stablecoin checkout rails

Part C: 12-Month Regional Expansion Horizon

Horizon Phase	Key Strategic Deliverables & Expansion Targets	Commercial Metric Target
Phase 1: Gauteng Dominance (Mo 1-3)	Anchor in Pretoria (O12), Braamfontein, Soweto & Mamelodi. Onboard 25 verified labels.	R500,000 Monthly GMV • 650 Orders/mo
Phase 2: Coastal Expansion (Mo 4-6)	Expand to Durban (Florida Rd / Umlazi) & Cape Town (Long St / Khayelitsha ateliers).	R1.5M Monthly GMV • 1,800 Orders/mo
Phase 3: Wholesale & B2B (Mo 7-9)	Launch Atelier Fabric & Wholesale Trim procurement portal for registered fashion designers.	R3.0M Monthly GMV • 3,500 Orders/mo

Chapter 10: Financial Impact, Unit Economics & ROI Model

Demonstrating sustainable marketplace unit economics, designer margin expansion, and platform cash flow.

"As Head of Product, my mission is to prove that supporting grassroots African culture is not charity—it is an immensely lucrative, mathematically unassailable commercial engine. By fixing last-mile delivery with smart lockers and lowering commissions from 30% to 13%, we create a flywheel where designers earn 3x more and Le Benkeleng builds an ultra-lean, highly profitable marketplace."

— Head of Product / Commercial Strategist

Representative Order Unit Economics (R950 Average Basket)

UNIT ECONOMIC DRIVER	TRADITIONAL DEPARTMENT STORE / LEGACY RETAIL	LE BENKELENG AUTONOMOUS MARKETPLACE	VARIANCE / BENEFIT
Gross Order Value (GMV)	R950.00 (Customer retail price)	R950.00 (Customer retail price)	Parity in retail pricing
Retailer Commission / Take	30.0% = R285.00 retained by retailer	13.0% = R123.50 retained by Le Benkeleng	R161.50 saved for the designer
Net Designer Payout	R665.00 (Paid after 60-day credit terms)	R826.50 (Paid automatically on locker delivery)	+24.3% higher net cash to designer
Last-Mile Delivery Cost	R140.00 (Suburban door courier; 42% failed)	R55.00 (Bob Go smart locker; 99.4% pickup)	R85.00 logistics savings per order
Payment Gateway Cost	R33.25 (3.5% credit card processing fee)	R23.75 (2.5% blended Capitec / Ozow EFT / Card)	R9.50 transaction fee savings
Net Platform Contribution	High overhead; physical lease & staff costs	R74.75 net contribution margin per order (60.5%)	Ultra-lean digital marketplace model

12-Month Commercial Scale & Projections

Marketplace Revenue Model

13% Marketplace Take-Rate: Levied on all designer apparel sales.
20% Thrift Vault Margin: Curated archive acquisition markup.
Atelier Studio SaaS (Tier 2): R350/mo for advanced multi-staff tools.
Locker Logistics Margin: R10 spread on negotiated courier rates.

Year 1 Projected GMV: R28,500,000 • Platform Revenue: R3,705,000



60.5%
NET PLATFORM CONTRIBUTION MARGIN ON 13% TAKE-RATE

Appendix: Technical Reference & Codebase Catalog

Exhaustive REST API endpoints, webhook contracts, legal disclosures, and domain glossary.

Core REST API Endpoints Catalog

METHOD & ENDPOINT	PAYLOAD / QUERY PARAMETERS	RESPONSE CONTRACT	SECURITY & DESCRIPTION
GET /api/v1/marketplace/catalog	?category=outerwear&city=pretoria	{products: [...], totalCount: N}	Public CDN cached; returns active products with live per-size stock
POST /api/v1/checkout/initialize	{items: [...], locker_station_id, payment_method}	{checkout_url, payment_ref, expires_in}	Generates 10-minute Redis reservation lock for 1-of-1 thrift pieces
POST /api/v1/webhooks/peach-payments	{payment_ref, status, signature, amount}	{success: true, order_number}	HMAC-SHA256 verified webhook; confirms order & notifies vendor
POST /api/v1/webhooks/bob-go	{waybill_number, status, locker_pin}	{status: "acknowledged"}	Triggers automated WhatsApp notification when locker is loaded
POST /api/v1/atelier/inventory/update	{product_id, size, new_qty}	{success: true, updated_stock: N}	Bearer token auth; synchronizes per-size quantity in real time
POST /api/v1/atelier/orders/fulfill	{order_id, action: "generate_waybill"}	{waybill_number, label_pdf_url}	Assigns courier barcode and updates status to dispatched_to_locker

Verified Streetwear Labels

```
// Active South African Independent Brands
• Lesupa Atelier (Pretoria 012 Bespoke Luxury)
• Mokasi Streetwear (Mamelodi Graphic Silhouettes)
• Galxboy Heritage (Gauteng Street Culture Icon)
• Soweto Threads (Vilakazi Raw Denim & Workwear)
• Braam District (Downtown JHB Boxy Cuts)
• Gusheshe Classics (BMW E30 Motorsport Tribute)
```

Statutory Disclosures (ECTA Sec 43)

Legal Entity: Le Benkeleng (Pty) Ltd • Reg: 2026/048219/07.
Registered Office: FHM Head Office, Pretoria, Gauteng, South Africa.
Information Officer: registered@lebenkeleng.co.za.
Returns Hub: Hatfield Central Distribution Hub, 1122 Burnett St, Pretoria.

Domain Glossary & South African Streetwear Terminology

Le Benkeleng: Sesotho/Sepedi vernacular for 'at the shop' / general dealer. • **012:** Pretoria telephonic dialing code, cultural badge of honor for capital city creators. • **Thrift Zone** ⚡: Curated 1-of-1 vintage and archive vault. • **Bob Go:** South African smart locker and courier logistics network. • **POPIA:** Protection of Personal Information Act 4 of 2013. • **CPA:** Consumer Protection Act 68 of 2008. • **Discreet Mode:** Privacy shield masking financial figures in public spaces.